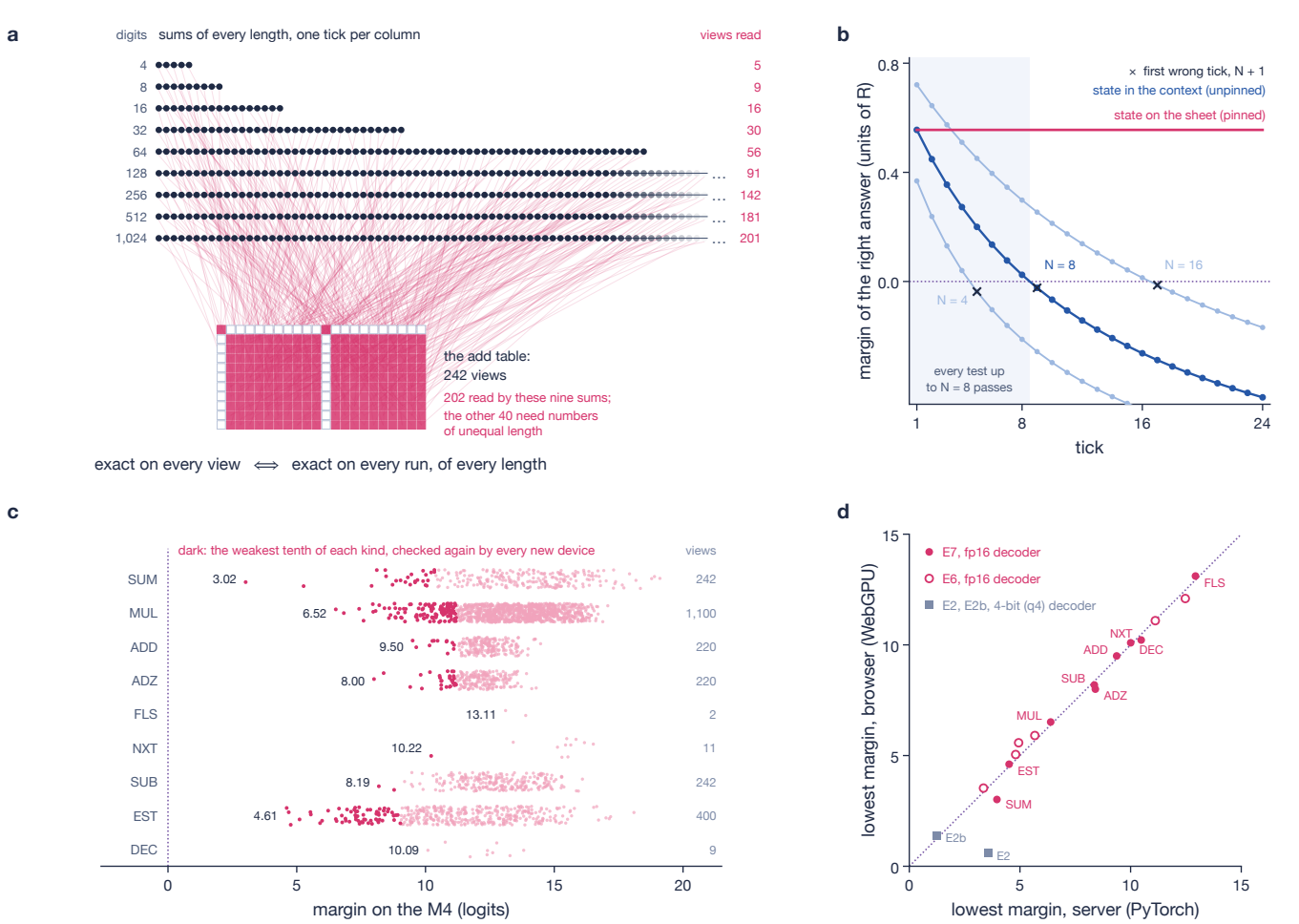

a
digits
sums of every length, one tick per column
views read
4
8
16
32
64
128
256
512
1,024
5
9
16
30
56
91
142
181
201
the add table:
242 views
202 read by these nine sums;
the other 40 need numbers
of unequal length
exact on every view ⟺ exact on every run, of every length
b
margin of the right answer (units of R)
× first wrong tick, N + 1
state in the context (unpinned)
state on the sheet (pinned)
N = 4
N = 8
N = 16
every test up
to N = 8 passes
tick
c
dark: the weakest tenth of each kind, checked again by every new device
views
SUM
MUL
ADD
ADZ
FLS
NXT
SUB
EST
DEC
3.02
6.52
9.50
8.00
13.11
10.22
8.19
4.61
10.09
242
1,100
220
220
2
11
242
400
9
margin on the M4 (logits)
d
E7, fp16 decoder
E6, fp16 decoder
E2, E2b, 4-bit (q4) decoder
lowest margin, browser (WebGPU)
lowest margin, server (PyTorch)
FLS
NXT
ADD
DEC
SUB
ADZ
MUL
EST
SUM
E2b
E2